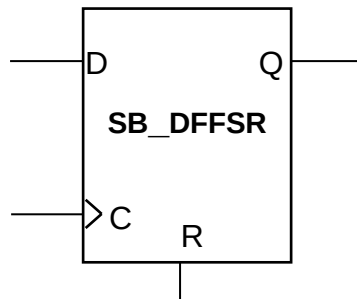


SB_DFFSR

D Flip-Flop with Synchronous Reset

Data: D is loaded into the flip-flop when Reset R is low during a rising clock edge transition.

Reset: R input is active high, overrides all other inputs and resets the Q output during a rising clock edge.



Inputs			Output
R	D	C	Q
1	X		0
X	X	0	No Change
0	0		0
0	1		1
Power on State	X	X	0

Key

Rising Edge
1 High logic level
0 Low logic level
X Don't care
? Unknown

HDL Usage

This register is inferred during synthesis and can also be explicitly instantiated.

Default Signal Values

The iCEcube2 software assigns the following signal values to unconnected input ports:

Input D: Logic '0'

Input C: Logic '0'

Input R: Logic '0'

Verilog Instantiation

```
// SB_DFFSR - D Flip-Flop, Reset is synchronous with the rising clock edge
```

```
SB_DFFSR SB_DFFSR_inst (  
    .Q(Q),           // Registered Output  
    .C(C),           // Clock  
    .D(D),           // Data  
    .R(R)            // Synchronous Reset  
);
```

```
// End of SB_DFFSR instantiation
```

VHDL Instantiation

-- SB_DFFSR - D Flip-Flop, Reset is synchronous with the rising clock edge

```
SB_DFFSR_inst : SB_DFFSR
  port map (
    Q => Q,          -- Registered Output
    C => C,          -- Clock
    D => D,          -- Data
    R => R           -- Synchronous Reset
  );

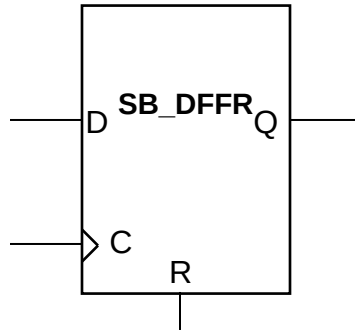
-- End of SB_DFFSR instantiation
```

SB_DFFR

D Flip-Flop with Asynchronous Reset

Data: D is loaded into the flip-flop when R is low during a rising clock edge transition.

Reset: R input is active high, overrides all other inputs and asynchronously resets the Q output.



Inputs			Output
R	D	C	Q
1	X	X	0
0	0	↗	0
0	1	↗	1
Power on State	X	X	0

Key

↗ Rising Edge
1 High logic level
0 Low logic level
X Don't care
? Unknown

HDL Usage

This register is inferred during synthesis and can also be explicitly instantiated.

Default Signal Values

The iCEcube2 software assigns the following signal values to unconnected input ports:

Input D: Logic '0'

Input C: Logic '0'

Input R: Logic '0'

Verilog Instantiation

// SB_DFFR - D Flip-Flop, Reset is asynchronous to the clock.

```
SB_DFFR    SB_DFFR_inst (
    .Q(Q),           // Registered Output
    .C(C),           // Clock
    .D(D),           // Data
    .R(R)            // Asynchronous Reset
);
```

```
// End of SB_DFFR instantiation
```

VHDL Instantiation

```
-- SB_DFFR - D Flip-Flop, Reset is asynchronous to the clock.
```

```
SB_DFFR_inst: SB_DFFR
  port map (
    Q => Q,          -- Registered Output
    C => C,          -- Clock
    D => D,          -- Data
    R => R           -- Asynchronous Reset
  );
```

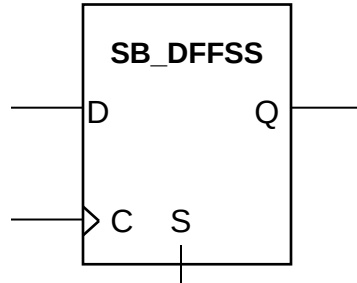
```
-- End of SB_DFFR instantiation
```

SB_DFFSS

D Flip-Flop with Synchronous Set

Data: D is loaded into the flip-flop when the Synchronous Set S is low during a rising clock edge transition.

Set: S input is active high, overrides all other inputs and synchronously sets the Q output.



Inputs			Output
S	D	C	Q
1	X		1
0	0		0
0	1		1
Power on State	X	X	0

Key

Rising Edge
1 High logic level
0 Low logic level
X Don't care
? Unknown

HDL Usage

This register is inferred during synthesis and can also be explicitly instantiated.

Default Signal Values

The iCEcube2 software assigns the following signal values to unconnected input ports:

Input D: Logic '0'

Input C: Logic '0'

Input S: Logic '0'

Verilog Instantiation

```
// SB_DFFSS - D Flip-Flop, Set is synchronous with the rising clock edge,
```

```
SB_DFFSS SB_DFFSS_inst (  
    .Q(Q),           // Registered output  
    .C(C),           // Clock  
    .D(D),           // Data  
    .S(S)            // Synchronous Set  
);
```

```
// End of SB_DFFSS instantiation
```

VHDL Instantiation

-- SB_DFFSS - D Flip-Flop, Set is synchronous with the rising clock edge

```
SB_DFFSS_inst  SB_DFFSS
  port map (
    Q => Q,          -- Registered Output
    C => C,          -- Clock
    D => D,          -- Data
    S => S           -- Synchronous Set
  );
```

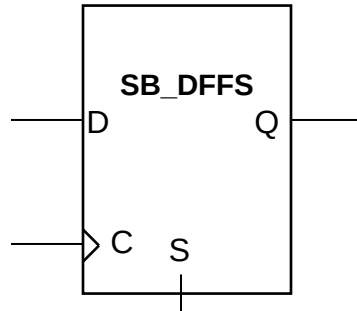
-- End of SB_DFFSS instantiation

SB_DFFS

D Flip-Flop with Asynchronous Set

Data: D is loaded into the flip-flop when S is low during a rising clock edge transition.

Set: S input is active high, and it overrides all other inputs and asynchronously sets the Q output.



Inputs			Output
S	D	C	Q
1	X	X	1
0	0	↗	0
0	1	↗	1
Power on State	X	X	0

Key

↗ Rising Edge
1 High logic level
0 Low logic level
X Don't care
? Unknown

HDL Usage

This register is inferred during synthesis and can also be explicitly instantiated.

Default Signal Values

The iCEcube2 software assigns the following signal values to unconnected input ports:

Input D: Logic '0'

Input C: Logic '0'

Input S: Logic '0'

Verilog Instantiation

// SB_DFFS - D Flip-Flop, Set is asynchronous to the rising clock edge

```
SB_DFFS    SB_DFFS_inst (
    .Q(Q),           // Registered Output
    .C(C),           // Clock
    .D(D),           // Data
    .S(S)            // Asynchronous Set
);
```

// End of SB_DFFS instantiation

VHDL Instantiation

-- SB_DFFS - D Flip-Flop, Set is asynchronous to the rising clock edge

```
SB_DFFS_inst: SB_DFFS
  port map (
    Q => Q,          -- Registered Output
    C => C,          -- Clock
    D => D,          -- Data
    S => S           -- Asynchronous Set
  );
```

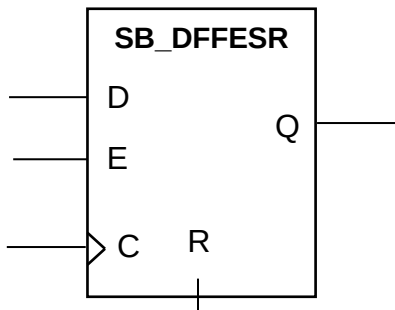
-- End of SB_DFFS instantiation

SB_DFFESR

D Flip-Flop with Clock Enable and Synchronous Reset

Data: D is loaded into the flip-flop when Reset R is low and Clock Enable E is high during a rising clock edge transition.

Reset: R, when asserted with Clock Enable E high, synchronously resets the Q output during a rising clock edge.



Inputs				Output
R	E	D	C	Q
1	1	X		0
X	0	X	X	Previous Q
0	1	0		0
0	1	1		1
Power on State	X	X	X	0

Key
 Rising Edge
1 High logic level
0 Low logic level
X Don't care
? Unknown

HDL Usage

This register is inferred during synthesis and can also be explicitly instantiated.

Default Signal Values

The iCEcube2 software assigns the following signal values to unconnected input ports:

- Input D: Logic '0'
- Input C: Logic '0'
- Input R: Logic '0'
- Input E: Logic '1'

Note that explicitly connecting a Logic '1' value to port E will result in a non-optimal implementation, since an extra LUT will be used to generate the Logic '1'. If the user's intention is to keep the FF always enabled, it is recommended that either port E be left unconnected, or the corresponding FF without a Clock Enable port be used.

Verilog Instantiation

```
// SB_DFFESR - D Flip-Flop, Reset is synchronous with rising clock edge
// Clock Enable.
```

```
SB_DFFESR SB_DFFESR_inst (
    .Q(Q),           // Registered Output
    .C(C),           // Clock
    .E(E),           // Clock Enable
    .D(D),           // Data
    .R(R)            // Synchronous Reset
);
```

```
// End of SB_DFFESR instantiation
```

VHDL Instantiation

```
-- SB_DFFESR - D Flip-Flop, Reset is synchronous with rising clock edge
-- Clock Enable.
```

```
SB_DFFESR_inst: SB_DFFESR
    port map (
        Q => Q,       -- Registered Output
        C => C,       -- Clock
        E => E,       -- Clock Enable
        D => D,       -- Data
        R => R        -- Synchronous Reset
    );
```

```
-- End of SB_DFFESR instantiation
```